

Precise timestamping and temporal synchronization in multi-sensor fusion

Authors: Tobias Huck, Antje Westenberger, Martin Fritzsche, Tilo Schwarz, Klaus C. J. Dietmayer (2011)

ABSTRACT

Multi-sensor fusion in robotics requires precise temporal alignment of data packets. The authors present a method for precise timestamping and synchronization between multiple sensor networks and central processing units, analyzing and mitigating network jitter and transmission latency.

RELEVANCE TO THE EB15 ROBOTIC ARM PROJECT

Provides theoretical support for the synchronization of the AS5600 encoder feedback readings between the ESP32 and Arduino.

TECHNICAL ANALYSIS & SYSTEM INTEGRATION

This academic research reference documents crucial design paradigms directly relevant to the development of the EB15 six-degrees-of-freedom robotic manipulator. Under the distributed control framework designed for this thesis (featuring the ESP32 S3 as master and the Arduino Uno as slave), the mathematical formulations, communication latencies, and performance profiles analyzed by Tobias Huck provide a benchmark for physical and algorithmic modeling. Specifically, this reference establishes solid validation criteria for timing constraints, closed-loop feedback via absolute magnetic encoders (AS5600), and vibration damping.